Buying Desk — The PO-Exception Agent for Retailers

Part 1 of 3 · The research behind the product choice

July 2026

1 · The short version

Over eight weeks (May to early July 2026) we ran a structured market-research programme to find the right first product for the venture. It was not a brainstorm — it was a repeatable pipeline, run twelve times, and it was designed to kill ideas, not to fall in love with them.

We searched for winnable ground, not exciting ground. We are a lean team, not a seed-funded startup, so we deliberately stayed out of the flashy AI categories — that is where startups that have raised $20–100M outspend each other for the same customers. Even if we built a superb product, it would not get far there: buyers hear from the companies with the biggest sales and marketing budgets, not from whoever built the best product — and a rival with ten times the engineers can copy any good feature within months.

Our advantage lies elsewhere: in problems that hurt just as much but get none of the attention — the unglamorous daily work the big players overlook. We will still have a strong go-to-market strategy — sales and marketing included — but the difference is what it takes to be heard. On crowded ground, being heard costs millions. On this ground, a focused outreach from a team that knows the customer's world deeply — and has their trust — wins a customer at a fraction of the cost.

The numbers:

- **12 discovery scans**, each looking at a different territory (retail operations, compliance, e-commerce, Reddit-sourced pains, and more)

- **60+ ideas** generated and scored against a 12-factor scorecard, with every competitor claim checked against live sources
- **8+ red-team reviews** on the shortlist — "red-teaming" an idea means switching sides and arguing against it, deliberately hunting for the evidence that kills it
- **Exactly one idea survived every stage:** the retailer-side **purchase-order exception agent**, which we've named **Buying Desk**

What the surviving idea is, in one paragraph. Mid-market retailers send thousands of purchase orders to suppliers. Most suppliers confirm by email and PDF, not by system-to-system EDI. So when an order goes wrong — no confirmation, a price that doesn't match the contract, a changed quantity or date, a short delivery — the ERP can't see it, and the buying team chases it by hand out of a shared mailbox. The misses cost real money (overpaid invoices, unclaimed penalties, empty shelves at a promotion launch), and most retailers cannot even put a number on the loss. Buying Desk is an AI agent that watches the orders and the supplier emails, catches these exceptions, does the chasing and the paperwork itself, and brings a human only the decisions that need real judgement. It never places orders and never makes buying decisions.

Why this document exists. This is Part 1 of a three-part series. **Part 0 — the strategy** — sets out where we choose to compete, how we win there, and how the venture is paid for. This document's job is to put the whole selection map on the table — including the ideas that looked better on paper and the evidence that ruled each one out — so we work from the same picture and don't spend future conversations re-opening ideas that have already failed on evidence. **Part 2** then sets out precisely what still has to be validated before we write production code, and how the work with Smyths, as our anchor design partner, answers each of those questions.

Where the product thinking already stands: a clickable prototype of the agent's workspace (the Buying Desk prototype, with light and dark mode), a plain-language product write-up, and an internal product requirements document with the kill-tests built in as gates. The prototype deliberately contains no invented performance numbers — the metrics it shows are the pass-lines we commit to measuring in a pilot, not claims.

- **Prototype** — the Buying Desk workspace, clickable
- **Product write-up** — the agent explained in plain language

2 · How we researched the market — the four-stage pipeline

Every idea went through the same four stages. The goal was that the surviving idea earns its place, rather than being anyone's favourite.

Stage 1 — Discovery. Generate candidate ideas systematically, seeded by a written thesis of what this venture is good at: a lean team, UK/US customers, AI agents that *do the work* rather than dashboards that show it, one problem solved to the core, and the retail domain knowledge we start with — a decade of building and running a real buying desk's tech solutions at Smyths, a relationship this venture must strengthen and never put at risk. Sources: vertical market maps, funding databases, regulation calendars and practitioner forums.

Stage 2 — Scoring. Each candidate scored on 12 factors (out of 60 raw): how sharp and how urgent the pain is, whether we can reach the buyer, what a customer would pay per year, how strong and how exposed the incumbents are, whether there is a defensible advantage ("moat" — the thing a competitor can't easily copy), integration depth, how often the problem occurs, regulation risk, fit with our team, and why *now* is the right time. Every competitive claim was checked against live web sources at scan time, not taken from memory.

Stage 3 — The red-team review. For each shortlisted idea, we switched sides and tried to kill it: hunted for funded competitors on the exact same wedge, free alternatives, bundling by bigger players, and structural problems. The biggest risks that survived were then turned into cheap tests we could run before building anything. Most of the shortlist died at this stage — including ideas that had outscored the eventual front-runner.

Stage 4 — Voice of the customer. Bottom-up mining of Reddit practitioner forums (r/supplychain, r/procurement, r/ERP, r/retail, r/smallbusiness and others, 16+ query batteries) to confirm the pain is expressed, recurring and quantified by real people in their own words — not just plausible to us.

A lesson we paid for and kept: Reddit tells you a pain is *felt*, not that it is *unsolved*. Loud pain attracts every builder, so the loudest threads often mark the most crowded categories. Every promising Reddit find was immediately checked against the incumbent landscape and a “could £15 of ChatGPT do this?” test. The prize is pain that is real **and** unsolved **and** worth real money **and** reachable. Almost everything hits only “real”.

3 · The funnel — 12 scans, 60+ ideas, one survivor

When	Scan	What it covered	Outcome
9 Jun	B2B ops landscape	~18 operations verticals (quoting, deductions, medical billing, credentialing, freight, tax…)	Every vertical already had an AI-native entrant. Calibrated the whole search: the opportunity is a defensible slice hidden behind friction, not open space.
23–25 Jun	Retail deep teardown	PO-exception vs stock-replenishment head-to-head, plus a sweep of the “full automation” quadrant (bookkeeping, KYC, claims…)	PO-exception scored 49/60 and beat replenishment (42/60 — the big incumbents RELEX and Blue Yonder have already shipped agentic replenishment). The automation quadrant: all taken or blocked.

When	Scan	What it covered	Outcome
26 Jun	10-idea open scan	Certified payroll, subcontractor compliance, licence renewals, shipping disputes, credentialing, insurance certificates, sales tax…	Certified payroll ranked #1 at 50/60 — it became the benchmark the PO idea had to beat. (It later died in the red-team review.)
late Jun	Top-25 slate	25 ideas across 12 verticals, with 5 deep-dive dossiers	Top tier: certified payroll 50 · PO-exception 48–49 · product-safety compliance 47 · three more at 46.
1 Jul	Retailer slates ×2	Two full top-10s built specifically for a Smyths-type retailer, plus 5 red-team reviews	The “obvious” retail money ideas (supplier income recovery, packaging fees, chargebacks, logistics) are all consolidated. PO-exception was the only idea to survive a head-on attack.

When	Scan	What it covered	Outcome
1–2 Jul	Reddit crawls ×3	Practitioner forums across supply chain, procurement, ERP, retail	The PO pain confirmed three times in practitioners' own words, including a £-quantified thread (Section 6). No better idea surfaced.
2 Jul	Reddit top-10 + red-team review of the top 3	10 fresh Reddit-sourced ideas scored against the standing leaders	The benchmarks held. The red-team review then cut certified payroll from 50 to ~44 and the best fresh find from 46 to ~40. PO-exception was left as the only clean survivor.

When	Scan	What it covered	Outcome
2 Jul	PO-exception competitor scan	Every findable AI purchase-order player, funding checked	Buy-side *manufacturing* is now crowded and funded. The *retail* buy-side is unclaimed. We cut our own idea's score to reflect this (49 → ~46–47) and attached three survive-only-if conditions — they are Gates 1–3 in Part 2.
2–3 Jul	Compliance slate	10 EU/AI-compliance ideas; the best (EUDR deforestation) fully attacked	All dead or parked. Key finding: Brussels deferred both the AI Act high-risk rules and EUDR — deadline-driven demand can be postponed by a vote; operational pain cannot.

When	Scan	What it covered	Outcome
3 Jul	E-commerce storefront scan	~20 candidates across 5 sub-domains	Zero passed. Shopify bundles AI for free, customer-service agents are a billion-dollar category, and the platforms own the new agent protocols. The most closed territory of all.
3 Jul	Challenger scan	Five niches never checked before (tariff classification, restoration, telecom expense, lease reconciliation, utility bills)	All five already claimed. **Nothing on the board beat PO-exception — so we paused discovery here and moved to validation.**

The base rate is the argument. Sixty-plus ideas, eight weeks, every fashionable territory checked — and exactly one clean survivor. Not because we were gentle with it: its score was cut twice as new competitive evidence arrived, and it carries three explicit conditions. It is #1 because everything else died faster. Discovery is paused, not finished: the search stopped because validating this candidate is now a better use of time than scanning further — and if the gates in Part 2 kill it, the search reopens with the map below as its head start.

4 · The strongest challengers — and why each one died

These rejections are as useful as the surviving idea: each row is a category we now know not to enter, and the reasons behind them became the standing filters in Section 5.

Idea	Peak score	What killed or capped it	Verdict
Certified-payroll agent (US public-works contractors) — was #1 at 50/60	50 → ~44	Two construction-payroll platforms raised $50M and $21M with this built in; QuickBooks now produces the key government form for free. Structurally: certified payroll is derived from payroll data, so the advantage always flows back to whoever holds the payroll system — and they can bundle it.	Deflated

Idea	Peak score	What killed or capped it	Verdict
Product-safety compliance agent (GPSR)	47	The "obligated producer" problem: for a multi-brand retailer, the legal duty and the budget mostly sit with the *brand or importer*, not the retailer. The pain isn't fully the retailer's to buy.	Downgraded
Healthcare-staffing credentialing agent — best fresh Reddit find	46 → ~40	The US nursing board gives licence monitoring away free; funded AI entrants arriving; the travel-nurse market shrank 12% in a year; and the incumbent price anchor is $99/month — a lifestyle-business ceiling.	Weak

Idea	Peak score	What killed or capped it	Verdict
Supplier income / deduction recovery — the “obvious” retail money idea	—	The most consolidated corner of retail tech. The recovery-audit firms work on a share of what they find (zero upfront cost to the retailer), which destroys the case for paying a subscription.	Killed
Packaging / EPR eco-fee compliance agent	strong on paper	Two discovery claims proved false when checked: incumbents already automate the core job, and again the retailer usually isn’t the legally obligated party. Filing is often already bundled into scheme memberships the retailer pays for.	Killed

Idea	Peak score	What killed or capped it	Verdict
EUDR deforestation due-diligence agent	42	The core check exists **free, twice** (an open-source tool from WRI/FAO plus a free validator); compliance is being bundled upstream into supplier invoices and certifier programmes; the deadline slipped again; self-serve competitors charge €29–79/month.	Killed

Idea	Peak score	What killed or capped it	Verdict
Cost-price-increase (CPI) management agent	—	Partially survived: clean customer, no direct incumbent. But it borders existing SAP/contract tooling, only happens a few times a year per supplier, and the accept/reject decision stays human. **Kept as Agent #2 of the same suite** — not a lead product.	Suite #2
Carrier-portal shipment-tracking agent — looked like a sibling of the PO idea	—	project44 spent $420M aggregating 230,000+ carriers and opened instant access to 80,000 of them. The "long tail" this idea would automate has already been aggregated by a giant.	Killed

Idea	Peak score	What killed or capped it	Verdict
Field-inspection report writer — the loudest pain found on Reddit	—	Already a saturated AI category (the leader is worth ~$2.5B and its AI report-writing is the industry default), and undercut from below by a £15/month ChatGPT subscription. The definitive proof that loud pain ≠ open market.	Killed
Medical billing denials / prior authorisation	—	The reference anti-pattern: funded incumbents on the exact wedge, heavy healthcare regulation for a lean offshore team, and 9–18-month hospital sales cycles. The mirror image of the PO idea on every axis.	Killed

Idea	Peak score	What killed or capped it	Verdict
The whole e-commerce storefront territory (~20 candidates)	—	Platform bundling (Shopify gives AI away), the hottest venture-funding theme, and platform-owned protocols.	0 of 20 passed

5 · The kill-patterns we learned

The red-team reviews produced eight reusable rules. Any future idea — including extensions of our own suite — gets tested against these first.

1. **The advantage flows to the system of record.** If your product is derived from data someone else holds (payroll, forecasting, freight networks), that player can always bundle you away. *Killed: certified payroll, replenishment, carrier tracking.*
2. **Free or bundled kills willingness to pay.** If a regulator, non-profit or incumbent gives the core job away free — or bundles it into a relationship the customer already pays for — the market is gone before you start. *Killed: EUDR, staffing credentialing, and it capped certified payroll.*
3. **Check who legally owns the problem.** In compliance ideas, the duty (and the budget) often sits with the brand or importer, not the retailer you can reach. *Killed: packaging EPR; downgraded: product safety.* The PO idea is immune — the order, the money and the ERP are unambiguously the retailer's own.
4. **Has someone already aggregated the long tail?** "Automate the long-tail manual document job" only works where nobody has spent $100M+ building a network over that long tail. Freight status: someone had — dead. Non-EDI supplier confirmations: nobody has (the big order-automation firms serve the *seller's* side) — alive.

5. **Deadline-driven demand can be postponed.** Brussels deferred two major regulations in eight months. A purchase order going wrong today cannot be deferred by a vote.
6. **Loud pain on forums usually means a crowded market**, because acute pain attracts every builder. Pain evidence only counts alongside a competitor scan.
7. **“No win, no fee” incumbents destroy subscription pricing.** Where recovery firms take a share of findings, a paid product fights a free alternative.
8. **Never build on a surface the platform owns.** Anywhere Shopify, Amazon or TikTok can make your product a toggle, they eventually will.

6 · What buyers actually say — the PO pain threads

Three separate Reddit crawls surfaced this exact pain in practitioners’ own words — unprompted, recurring, and in one case with a £ figure attached. These quotes are also how we open the discovery conversations with the Smyths buying team (Part 2): “does this happen here?” — not a pitch.

> “Tired of constantly chasing suppliers for PO updates… the PO goes out, the supplier confirms, then the ship date moves again.” — r/supplychain (a practitioner describing the exception loop end-to-end)

> Teams still manually chasing supplier PO confirmations *after* an ERP implementation — “supplier confirmations not coming through the system, so people revert to chasing manually” — with the poster putting the extra cost at **£100k+**. — r/ERP (the non-EDI failure mode, with a rare £ figure on it)

> Asked what they would automate first, procurement practitioners’ #1 answer: **“vendor follow-ups.”** — r/procurement

> “Manual it is. When they get a purchase confirmation from a vendor (usually a PDF via email), they have to sit down and manually type every line.” — r/smallbusiness (a store owner, thread 1nz4mb4)

> “Why is sales order processing still so manual in 2025?” — r/manufacturing (the seller’s side of the same broken paperwork — confirming the problem exists on both ends of the PO)

One caution the crawls also produced: the low end of this pain is being DIY-automated right now — "PDF order into ERP, zero typing" threads using off-the-shelf automation tools are appearing. Raw document extraction is becoming a commodity. That sharpened our definition of the defensible layer: **it is the retail reconciliation logic, the exception-resolution workflow, the approved write-path into SAP, and the memory of how each supplier's exceptions actually get resolved — never the PDF parsing.**

7 · The front-runner — why the PO-exception agent survived every test

Why it survived when 60+ ideas didn't:

- **It passes every kill-pattern in Section 5.** The retailer owns the pain, the money and the ERP. Nobody has aggregated the non-EDI supplier long tail — the big order-automation incumbents (Conexiom, Esker) work for the *seller*, automating incoming customer orders; a retailer checking a received confirmation against its own PO is the mirror-image job they don't do. Nothing about it is free, bundled, deferrable or platform-owned.
- **High frequency, low judgement.** Exceptions arrive daily (not quarterly or annually), and most fixes are paperwork rather than commercial decisions — exactly what a buyer will delegate to an agent, and exactly where agents are strong.
- **A genuine "why now".** Connecting a supplier by EDI stops being worth the cost at roughly 40–50 suppliers — that's why the long tail exists. Large language models have just made reading that long tail of messy emailed confirmations economic for the first time. This is the best-timed idea on the whole board.
- **Demand is proven with other people's money.** The equivalent product for *manufacturers* attracted over $100M of funding in 18 months, and ERP vendors are already buying those startups (Kavida → QAD; Traverse → SPS Commerce). Buyers pay for this job — and the retail slice is still unclaimed.

- **It fits us specifically.** The buying cockpit already running inside Smyths is the exact surface this agent needs — the approved SAP write-path, the buyers who already trust the tool. For an outsider this is a cold, slow, trust-gated sale; for us it is a warm conversation with a live anchor customer. The idea and the partnership select each other.

The score history — including the markdowns we made as new evidence arrived:

Date	Score (raw / 60)	What moved it
25 Jun	**49** — Strong	Deep verification: pain, price point (~£35k/yr), and buyer access all confirmed.
1 Jul	**49 held** — survived a head-on attack	The obvious kill ("EDI already solves this") fails: EDI covers only ~15–30% of supplier volume. The real risk moved to **distribution** — can we reach the buyer?
2 Jul	**~46–47** after the competitor scan	The manufacturing version of this product is crowded and funded, so we cut our own score. Rank unchanged — everything else fell further.

Date	Score (raw / 60)	What moved it
3 Jul	**Still #1 — discovery paused**	A further challenger scan found nothing better, so scanning stopped in favour of validation. Verdict: survives, subject to the three conditions now formalised as Gates 1–3 in Part 2.

The suite behind the first product. Several ideas that died as standalone bets survive as *later agents on the same desk, for the same buyer, through the same connector*: the cost-price-increase agent (#2), the article/SKU-setup agent (#3 — thousands of seasonal new products, heavy data entry), supplier onboarding, claims and rebates. We sell ONE agent first; the suite is the architecture, not the pitch. And one boundary is now fixed across every document: **the product does the work; it never makes the buying decisions.** No forecasting, no replenishment, no assortment calls — that keeps us out of the big incumbents' territory and inside the buyer's trust.

8 · The competition — the full map

Strongest threats first, including the ones that shaped our positioning.

Player	What they are	Why they don't (currently) own our slice
Didero ($30M raised, Feb 2026, Microsoft's fund)	AI purchase-order agents on top of existing ERP and email — the closest funded analogue	Serves manufacturers and distributors (~30 customers, no retail names found). But this sets the clock: realistically 12–18 months before funded players cross into retail.

Player	What they are	Why they don't (currently) own our slice
SourceDay (~$70M raised)	The incumbent PO-collaboration platform, now adding AI exception agents	Supplier-portal model, manufacturing DNA, sold through a manufacturing ERP channel.
Kavida → QAD (acquired Nov 2025)	AI post-PO management	Absorbed into a manufacturing ERP. Also the exit comparison: ERP vendors buy these companies.
Conexiom / Esker	The big order-automation incumbents (billions of PO lines)	They serve the *seller's* side. Our job is their mirror image. Worth watching for a pivot.
SPS Commerce "MAX"	AI over 300,000+ retail trading relationships, surfacing PO/shipping exceptions	The structural threat — but only for exceptions **visible on their network**. That is exactly why Gate 3 (Part 2) exists: our slice must be the exceptions trapped inside the retailer's own customised SAP estate and email inbox, which no network can see.

Player	What they are	Why they don't (currently) own our slice
SAP itself (Joule)	The platform incumbent	Its retail agents focus on assortment; its procurement agents live on the network side. It is weakest precisely in heavily customised estates — which is where we live.
DIY from below (workflow tools + OCR)	Commoditising raw document extraction	Can read a PDF; cannot do retail reconciliation logic, resolution workflow, an approved SAP write-path, or accumulate supplier-behaviour memory. That layer is the product.

The strategic consequence: the positioning has been narrowed twice and is now precise — **the exceptions trapped inside customised SAP retail estates, reached through the buying team's existing cockpit.** Not "AI for procurement". Generic positioning now loses to funded players; the narrow position is defensible and ours to take first. Speed matters — the same scan that proved the demand also started the clock.

Where this leads: the research says the idea is right; it does not yet say the business works. [Part 2 — The validation plan] turns the remaining risk into five gates with pass and kill lines, sets out the Smyths design-partner engagement in full, and lays out the 90-day programme.

Prepared July 2026. Compiled from the June–July 2026 research corpus: 12 discovery scans, 12-factor scorecards, 8+ red-team reports, three Reddit voice-of-customer crawls, and funding/competitor verification against live *web sources at the time of each scan. Competitive facts (funding rounds, acquisitions, product launches) were verified at scan time and should be re-confirmed before external use.*